

Nickel-Catalyzed Csp²–Csp³ Bond Formation via C–F Bond Activation

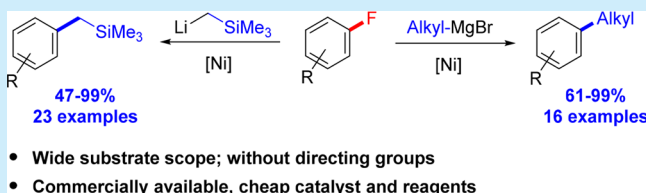
Yee Ann Ho,[†] Matthias Leiendecker,[†] Xiangqian Liu,[†] Chengming Wang,[†] Nurtalya Alandini,[†] and Magnus Rueping^{*,†,‡,§}

[†]Institute of Organic Chemistry, RWTH Aachen University, Landoltweg 1, D-52074 Aachen, Germany

[‡]KAUST Catalysis Center, King Abdullah University of Science and Technology (KAUST), Thuwal 23955-6900, Saudi Arabia

S Supporting Information

ABSTRACT: A nickel-catalyzed cross coupling of aryl fluorides via C–F bond activation has been developed. The alkylation method allows selective replacement of aryl fluorides by alkyl groups and enables the synthesis of diverse and otherwise difficult to access scaffolds in good yields.



In recent years, nickel-catalyzed cross-coupling reactions have received increasing attention as an alternative pathway for carbon–carbon bond construction due to the notable advantages of Ni catalysis in terms of economy and versatility. Being a relatively electropositive late-transition metal, nickel facilitates oxidative addition in cross-coupling reactions¹ and, therefore, enables cross-coupling reactions of electrophiles that are otherwise considerably less reactive under Pd catalysis. Thus, phenol derivatives,² aromatic nitriles,³ and aryl fluorides⁴ have been employed in nickel catalysis. Successful application of more inert bonds, such as C–F bonds, in cross-coupling reactions will allow functional group modification at a later stage for compounds containing additional functional moieties.⁵ Fluorinated arenes have received great interest due to their roles in drug discovery and materials science.⁶ Considering the intrinsic strength of the C–F bonds, the activation of unreactive Csp²–F bonds for further chemical transformations has motivated researchers to attempt overcoming this challenge and develop methodologies for Csp²–F bond activation. As a result, Csp²–F bond activation has been achieved, and successful protocols have evolved for Kumada,⁷ Suzuki,⁸ Negishi,⁹ Stille¹⁰ and Sonogashira¹¹ coupling reactions. While many examples were described for Csp²–Csp² and Csp²–Csp bond formations, only a few reports on Csp²–Csp³ bond formation exist. The first example was reported by Kumada et al. in 1973¹² in which the desired cross-coupling product had been obtained in low yield due to the formation of side products arising from β -hydride elimination. Improvement has been achieved by changing the catalyst to Ni(dppp)Cl₂. However, β -hydride elimination was unavoidable.¹³ Further progress was achieved by Nakamura et al. in 2012 with the introduction of a diphosphine POP ligand and use of in situ formed organozinc reagents.^{9f} Other precedents of Csp²–Csp³ bond formation from aryl fluoride are rare, usually taking advantage of an *ortho*-directing electron-withdrawing group, leading to a narrow scope for the alkylation of aryl fluorides.^{9a,14} The main challenge that

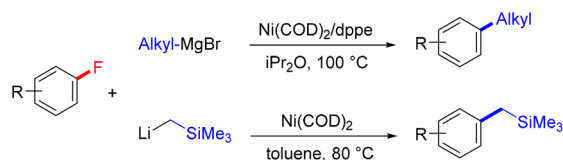
needs to be addressed in nickel-catalyzed C–F bond alkylations is suppression of the undesired β -hydride elimination. Herein, we report an efficient Ni-catalyzed alkylation of aryl fluorides which is suitable for a wide range of substrates, without the requirement of directing or activating groups. In addition to the coupling of aryl fluorides with Grignard reagents, the use of a Li nucleophile (LiCH₂SiMe₃) was also realized (Scheme 1).¹⁵

Regarding the choice of catalyst for the alkylation reaction, we started with the following considerations: (1) the bidentate structure of the ligand should stabilize the Ar–Ni^{II}–F complex formed; (2) COD (cyclooctadiene) introduced into the reaction mixture could help in stabilizing the regenerated Ni⁰ until its next usage in a new catalytic cycle. As such, a catalytic system consisting of Ni(COD)₂ and readily available dppe was set as our starting point. The cross-coupling between commercially available **1a**, as aryl fluoride representative, and Grignard reagent **2a** was chosen as model reaction for the exploration of the optimal conditions. To our delight, the Ni(COD)₂/dppe combination proved to be effective, delivering the desired product in a high yield of 85% without the formation of β -hydride elimination side product (Table 1, entry 1). This result could be further improved to an excellent yield of 99% after increasing the reaction temperature to 100 °C (entry 2). However, when the reaction temperature was raised to 120 °C, the yield of the desired product **3a** decreased to 84% (entry 3). Further exploration of other ligands was also carried out. A control reaction, in which no added ligand was used, gave the product in a poor yield of 12% (entry 4), stressing the importance of an additional ligand in the reaction. Bidentate diphosphine ligands such as dcype and dppf provided good yields of 96% and 87% (entries 5 and 6). On the other hand, monodentate phosphine ligands PPh₃, PCy₃,

Received: July 25, 2018

Published: August 31, 2018

Scheme 1. Nickel-Catalyzed C–F Bond Alkylation



- Wide substrate scope
- Excellent yields (up to 99% yield)
- Without directing groups
- Cheap, commercially available catalyst and reagents

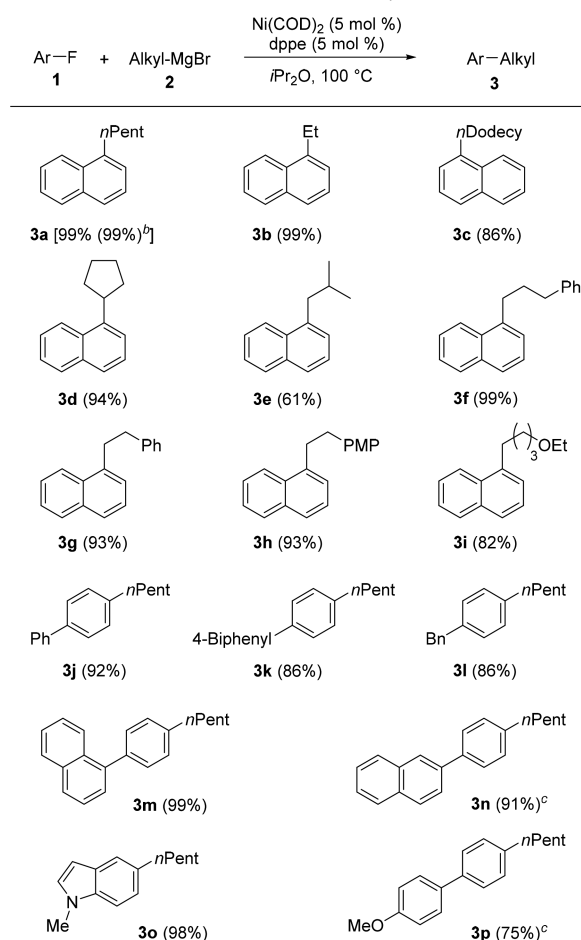
Table 1. Optimization of the Reaction Conditions^a

entry	ligand	temp (°C)	solvent	yield ^b (%)
1	dppe	80	<i>i</i> Pr ₂ O	85
2	dppe	100	<i>i</i> Pr ₂ O	99
3	dppe	120	<i>i</i> Pr ₂ O	84
4	—	100	<i>i</i> Pr ₂ O	12
5	dcype	100	<i>i</i> Pr ₂ O	96
6	dppf	100	<i>i</i> Pr ₂ O	87
7 ^c	PPh ₃	100	<i>i</i> Pr ₂ O	60 ^d
8 ^c	PCy ₃	100	<i>i</i> Pr ₂ O	33
9 ^c	XPhos	100	<i>i</i> Pr ₂ O	12
10	dppe	100	Et ₂ O	88
11	dppe	100	toluene	96

^aReaction conditions: **1a** (0.25 mmol), **2a** (0.5 mmol), Ni(COD)₂ (5 mol %), ligand (5 mol %), solvent (1 mL), 24 h. ^bYield after purification. ^c10 mol % ligand was used. ^dUndesired homocoupling side product was formed as well.

and XPhos resulted in poor yields, with large amounts of unreacted starting material or unwanted homocoupling side product (entries 7–9). The bidentate diphosphine ligands provide sufficient rigidity compared to the monodentate phosphine ligands and could stabilize the Ar–Ni^{II}–F complex formed after oxidative addition.¹⁶ The use of readily available and air-stable dppe is favorable since it provided the desired product in 99% yield. Next, different solvents were evaluated (entries 2, 10, and 11), and *i*Pr₂O proved to be the solvent of choice (entry 2).

With the optimized reaction conditions in hand, the scope of the reaction with respect to the structure of both coupling partners was evaluated. Notably, a wide range of fluorinated substrates and different types of alkyl Grignard reagents could be applied in the reaction, providing the corresponding alkylated products **3a–p** with excellent yields (Scheme 2). For example, the Csp²–Csp³ bond-forming reaction of aryl fluoride **1a** was successful with different Grignard reagents **2a–i**, leading to products **3a–i** in up to 99% yield. In addition to the *n*-pentyl derivative **2a**, Grignard reagents bearing short as well as long alkyl chain groups such as ethyl (**2b**) and *n*-dodecyl (**2c**) also performed well in this transformation. Notably, secondary, cyclic Grignard reagent **2d** and branched alkyl derivative **2e** were well tolerated, providing products **3d** and **3e** in 94% and 61% yields, respectively. Furthermore, the reaction was also successful with substrates **2f–h** that bear an aromatic group at the terminal position, affording the coupling products **3f–h** in excellent yields (93–99%). Our reaction

Scheme 2. Substrate Scope of Ni-Catalyzed Csp²–Csp³ Bond Formation via Activation of Aryl Fluorides^a

^aReaction conditions: **1a** (0.25 mmol), **2a** (0.5 mmol), Ni(COD)₂ (5 mol %), dppe (5 mol %), *i*Pr₂O (1 mL), 100 °C, 24–60 h, yields after purification. ^bReaction on 1 mmol scale, 72 h. ^c10 mol % of Ni(COD)₂/dppe.

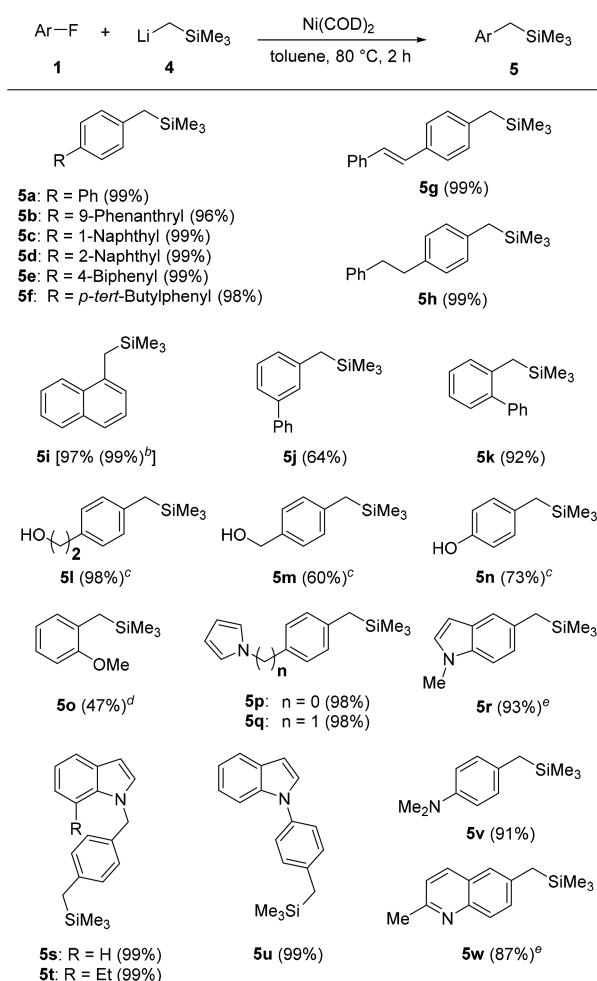
Table 2. Optimization of the Reaction Conditions^a

entry	Ni(COD) ₂ (X mol %)	temp (°C)	solvent	yield ^b (%)
1	2.5	rt	toluene	43
2	2.5	50	toluene	75
3	2.5	80	toluene	99
4	2.5	60	THF	47
5	2.5	30	Et ₂ O	60
6	1	80	toluene	92
7	0.5	80	toluene	42
8	—	80	toluene	—

^aReaction conditions: **1b** (0.25 mmol), **4** (1.3 equiv), Ni(COD)₂ (X mol %), solvent (1.5 mL), 2 h. ^bYield after purification.

conditions were also efficient when an ether functional group was present at the terminal position of the alkyl chain (**3i**, 82% yield). Next, the scope of the reaction with respect to the structure of the fluorinated coupling partner was evaluated. Substrates **1b–f**, bearing various aryl substituents in the *para*-position of the fluorobenzene ring including phenyl (**1b**),

Scheme 3. Substrate Scope of the Aryl Fluoride Functionalization Reaction^a



^aReaction conditions: **1** (0.25 mmol), **4** (1.3 equiv), Ni(COD)_2 (2.5 mol %), toluene (1.5 mL), 2 h, yields after purification. ^bReaction on 1 mmol scale, 2 h. ^c3 equiv of $\text{LiCH}_2\text{SiMe}_3$, 5 mol % of Ni(COD)_2 , 16 h. ^d1.1 equiv of $\text{LiCH}_2\text{SiMe}_3$. ^e5 mol % of Ni(COD)_2 , 16 h reaction time.

biphenyl (**1c**), benzyl (**1d**), and naphthyl (**1e**, and **1f**) led to full conversion of the starting materials and high yields (86–99%). Indole derivative **1g** underwent the reaction smoothly, providing the desired product **3o** with an excellent yield of 98%. For substrate **1h** that bears a methoxy functional group, the methoxy group successfully survived the reaction conditions furnishing the desired product **3p** in 75% yield.

Next, our attention was drawn to the commercially available nucleophile $\text{LiCH}_2\text{SiMe}_3$. We started the optimization study by reacting 4-biphenyl fluoride (**1b**) with $\text{LiCH}_2\text{SiMe}_3$ (**4**) in the presence of 2.5 mol % of Ni(COD)_2 in toluene at room temperature. Pleasingly, we obtained the corresponding $\text{ArCH}_2\text{SiMe}_3$ **5a** product in 43% yield (Table 2, entry 1). The yield could be considerably increased by raising the temperature to 80°C (99%, entry 3). Lower yields were obtained when the reaction was carried out in THF (47%, entry 4) and diethyl ether (60%, entry 5). Reducing the catalyst loading to 1 mol % resulted in a slightly lower yield of 92% (entry 6), whereas a further decrease to 0.5 mol % led to a significantly lower yield of 42% (entry 7). Furthermore, in the

absence of the Ni catalyst, no conversion of the starting material was observed (entry 8).

This $\text{Csp}^2\text{--Csp}^3$ coupling reaction could be applied to a range of fluorinated substrates with generally excellent yields (Scheme 3). Various aryl substituents in the *para*-position of fluorobenzene led to full conversion of the starting materials and high yields (96–99%) for the desired products **5a–f**. The *trans*-stilbene substrate and its saturated derivative could be converted to **5g** and **5h** in quantitative yields. In addition, the 1-naphthyl derivative **5i** was obtained in 97% yield. Pleasingly, various alcohol derivatives were found to be good substrates for the coupling reaction, which produced good to high yields for the aromatic (**5n**), benzylic (**5m**), and aliphatic (**5l**) derivatives. A series of nitrogen-containing compounds could also efficiently be subjected to the reaction conditions, yielding products **5p–w** with high to excellent yields (87–99%).

In summary, a direct method for the $\text{Csp}^2\text{--Csp}^3$ bond formation with aryl fluorides as electrophiles has been developed. By using a $\text{Ni(COD)}_2/\text{dppe}$ catalytic system, the alkylation of aryl fluorides could be performed with various alkyl Grignard reagents with yields of up to 99%. The established methodology proved successful in avoiding the undesired β -hydride elimination and is appealing in terms of electrophile availability as no directing or activating groups are required. Moreover, the components of the catalytic system (Ni(COD)_2 and dppe) and the Grignard reagents are readily available. In addition, the use of $\text{LiCH}_2\text{SiMe}_3$ as nucleophile resulted in products that can be converted into more complex structures by taking advantage of the existing SiMe_3 group.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.orglett.8b02351.

Detailed experimental procedures, spectral data for all compounds, and copies of ^1H , ^{13}C , and ^{19}F NMR spectra (PDF)

■ AUTHOR INFORMATION

Corresponding Author

*E-mail: magnus.rueping@kaust.edu.sa, magnus.rueping@rwth-aachen.de.

ORCID

Magnus Rueping: 0000-0003-4580-5227

Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

Y.A.H. acknowledges the DAAD for a scholarship. M.L. was supported by a Kekulé fellowship (Fonds der Chemischen Industrie) and the Studienstiftung des Deutschen Volkes.

■ REFERENCES

- (1) Tsou, T. T.; Kochi, J. K. *J. Am. Chem. Soc.* **1979**, *101*, 6319.
- (2) (a) Li, B.-J.; Yu, D.-G.; Sun, C.-L.; Shi, Z.-J. *Chem. - Eur. J.* **2011**, *17*, 1728. (b) Mesganaw, T.; Garg, N. K. *Org. Process Res. Dev.* **2013**, *17*, 29. (c) Tobisu, M.; Takahira, T.; Chatani, N. *Org. Lett.* **2015**, *17*, 4352. (d) Nakamura, K.; Tobisu, M.; Chatani, N. *Org. Lett.* **2015**, *17*, 6142. (e) Tobisu, M.; Chatani, N. *Acc. Chem. Res.* **2015**, *48*, 1717. (f) Guo, L.; Liu, X.; Baumann, C.; Rueping, M. *Angew. Chem., Int. Ed.*

- 2016, 55, 15415. (g) Guo, L.; Hsiao, C.-C.; Yue, H.; Liu, X.; Rueping, M. *ACS Catal.* **2016**, 6, 4438. (h) Liu, X.; Hsiao, C.-C.; Kalvet, I.; Leiendecker, M.; Guo, L.; Schoenebeck, F.; Rueping, M. *Angew. Chem., Int. Ed.* **2016**, 55, 6093.
- (3) For early examples, see: (a) Miller, J. A. *Tetrahedron Lett.* **2001**, 42, 6991. (b) Garcia, J. J.; Brunkan, N. M.; Jones, W. D. *J. Am. Chem. Soc.* **2002**, 124, 9547. (c) Miller, J. A.; Dankwardt, J. W. *Tetrahedron Lett.* **2003**, 44, 1907.
- (4) LaBerge, N. A.; Love, J. A. In *Organometallic Fluorine Chemistry*; Topics in Organometallic Chemistry; Springer International Publishing, 2015; Vol. 52, pp 55–111.
- (5) (a) Kiplinger, J. L.; Richmond, T. G.; Osterberg, C. E. *Chem. Rev.* **1994**, 94, 373. (b) Amii, H.; Uneyama, K. *Chem. Rev.* **2009**, 109, 2119. (c) Sun, A. D.; Love, J. A. *Dalton Trans.* **2010**, 39, 10362. (d) Clot, E.; Eisenstein, O.; Jasim, N.; Macgregor, S. A.; McGrady, J. E.; Perutz, R. N. *Acc. Chem. Res.* **2011**, 44, 333. (e) Keyes, L.; Love, J. A. *RSC Catalysis Series* **2013**, 11, 159. (f) Ahrens, T.; Kohlmann, J.; Ahrens, M.; Braun, T. *Chem. Rev.* **2015**, 115, 931. (g) Unzner, T. A.; Magauer, T. *Tetrahedron Lett.* **2015**, 56, 877. (h) Shen, Q.; Huang, Y.-G.; Liu, C.; Xiao, J.-C.; Chen, Q.-Y.; Guo, Y. *J. Fluorine Chem.* **2015**, 179, 14.
- (6) (a) Meyer, E. A.; Castellano, R. K.; Diederich, F. *Angew. Chem., Int. Ed.* **2003**, 42, 1210. (b) Müller, K.; Faeh, C.; Diederich, F. *Science* **2007**, 317, 1881. (c) Murphy, A. R.; Fréchet, J. M. J. *Chem. Rev.* **2007**, 107, 1066. (d) Babudri, F.; Farinola, G. M.; Naso, F.; Ragni, R. *Chem. Commun.* **2007**, 1003. (e) Preshlock, S.; Tredwell, M.; Gouverneur, V. *Chem. Rev.* **2016**, 116, 719.
- (7) (a) Böhm, V. P. W.; Gstottmayr, C. W. K.; Weskamp, T.; Herrmann, W. A. *Angew. Chem., Int. Ed.* **2001**, 40, 3387. (b) Mongin, F.; Mojovic, L.; Guillet, B.; Trecourt, F.; Quéguiner, G. *J. Org. Chem.* **2002**, 67, 8991. (c) Dankwardt, J. W. *J. Organomet. Chem.* **2005**, 690, 932. (d) Ackermann, L.; Born, R.; Spatz, J. H.; Meyer, D. *Angew. Chem., Int. Ed.* **2005**, 44, 7216. (e) Yoshikai, N.; Mashima, H.; Nakamura, E. *J. Am. Chem. Soc.* **2005**, 127, 17978. (f) Saeki, T.; Takashima, Y.; Tamao, K. *Synlett* **2005**, 1771. (g) Ackermann, L.; Born, R.; Spatz, J. H.; Althammer, A.; Gschrei, C. J. *Pure Appl. Chem.* **2006**, 78, 209. (h) Korn, T. J.; Schade, M. A.; Cheemala, M. N.; Wirth, S.; Guevara, S. A.; Cahiez, G.; Knochel, P. *Synthesis* **2006**, 2006, 3547. (i) Korn, T. J.; Schade, M. A.; Wirth, S.; Knochel, P. *Org. Lett.* **2006**, 8, 725. (j) Inamoto, K.; Kuroda, J.; Sakamoto, T.; Hiroya, K. *Synthesis* **2007**, 2007, 2853. (k) Ramnial, T.; Taylor, S. A.; Bender, M. L.; Gorodetsky, B.; Lee, P. T. K.; Dickie, D. A.; McCollum, B. M.; Pye, C. C.; Walsby, C. J.; Clyburne, J. A. C. *J. Org. Chem.* **2008**, 73, 801. (l) Guan, B.-T.; Xiang, S.-K.; Wu, T.; Sun, Z.-P.; Wang, B.-Q.; Zhao, K.-Q.; Shi, Z.-J. *Chem. Commun.* **2008**, 1437. (m) Yoshikai, N.; Matsuda, H.; Nakamura, E. *J. Am. Chem. Soc.* **2009**, 131, 9590. (n) Wang, J.-R.; Manabe, K. *Org. Lett.* **2009**, 11, 741. (o) Xie, L.-G.; Wang, Z.-X. *Chem. - Eur. J.* **2010**, 16, 10332. (p) Ackermann, L.; Wechsler, C.; Kapdi, A. R.; Althammer, A. *Synlett* **2010**, 2010, 294. (q) Jin, Z.; Li, Y.-J.; Ma, Y.-Q.; Qiu, L.-L.; Fang, J.-X. *Chem. - Eur. J.* **2012**, 18, 446. (r) Guo, W.-J.; Wang, Z.-X. *J. Org. Chem.* **2013**, 78, 1054. (s) Shi, H.; Dai, W.; Wang, B.; Cao, S. *Organometallics* **2018**, 37, 459.
- (8) (a) Widdowson, D. A.; Wilhelm, R. *Chem. Commun.* **1999**, 2211. (b) Wilhelm, R.; Widdowson, D. A. *J. Chem. Soc. Perkin Trans. 1* **2000**, 3808. (c) Kim, Y. M.; Yu, S. J. *J. Am. Chem. Soc.* **2003**, 125, 1696. (d) Widdowson, D. A.; Wilhelm, R. *Chem. Commun.* **2003**, 578. (e) Mikami, K.; Miyamoto, T.; Hatano, M. *Chem. Commun.* **2004**, 2082. (f) Steffen, A.; Sladek, M. I.; Braun, T.; Neumann, B.; Stammler, H.-G. *Organometallics* **2005**, 24, 4057. (g) Liu, J.; Robins, M. J. *Org. Lett.* **2005**, 7, 1149. (h) Bahmanyar, S.; Borer, B. C.; Kim, Y. M.; Kurtz, D. M. *Org. Lett.* **2005**, 7, 1011. (i) Schaub, T.; Backes, M.; Radius, U. *J. Am. Chem. Soc.* **2006**, 128, 15964. (j) Ruiz, J. R.; Jimenez-Sanchidrian, C.; Mora, M. *J. Fluorine Chem.* **2006**, 127, 443. (k) Gallon, B. J.; Kojima, R. W.; Kaner, R. B.; Diaconescu, P. L. *Angew. Chem., Int. Ed.* **2007**, 46, 7251. (l) Cargill, M. R.; Sandford, G.; Tadeusiak, A. J.; Yufit, D. S.; Howard, J. A. K.; Kilickiran, P.; Nelles, G. *J. Org. Chem.* **2010**, 75, 5860. (m) Sun, A. D.; Love, J. A. *Org. Lett.* **2011**, 13, 2750. (n) Tobisu, M.; Xu, T.; Shimasaki, T.; Chatani, N. *J. Am. Chem. Soc.* **2011**, 133, 19505. (o) Kawamoto, K.; Kochi, T.; Sato, M.; Mizushima, E.; Kakiuchi, F. *Tetrahedron Lett.* **2011**, 52, 5888. (p) Yu, P.; Shen, Q.; Lu, L. *J. Org. Chem.* **2012**, 77, 1798. (q) Liu, X.-W.; Echavarren, J.; Zarate, C.; Martin, R. *J. Am. Chem. Soc.* **2015**, 137, 12470. (r) Niwa, T.; Ochiai, H.; Watanabe, Y.; Hosoya, T. *J. Am. Chem. Soc.* **2015**, 137, 14313. (s) Zhou, J.; Berthel, J. H. J.; Kuntze-Fechner, M. W.; Friedrich, A.; Marder, T. B.; Radius, U. *J. Org. Chem.* **2016**, 81, 5789.
- (9) (a) Wang, T.; Alfonso, B. J.; Love, J. A. *Org. Lett.* **2007**, 9, 5629. (b) Wang, T.; Love, J. A. *Organometallics* **2008**, 27, 3290. (c) Buckley, H. L.; Sun, A. D.; Love, J. A. *Organometallics* **2009**, 28, 6622. (d) Sun, A. D.; Love, J. A. *J. Fluorine Chem.* **2010**, 131, 1237. (e) Keyes, L.; Sun, A. D.; Love, J. A. *Eur. J. Org. Chem.* **2011**, 2011, 3985. (f) Nakamura, Y.; Yoshikai, N.; Ilies, L.; Nakamura, E. *Org. Lett.* **2012**, 14, 3316. (g) Wang, T.; Keyes, L.; Patrick, B. O.; Love, J. A. *Organometallics* **2012**, 31, 1397. (h) Zhu, F.; Wang, Z.-X. *J. Org. Chem.* **2014**, 79, 4285. (i) Asako, S.; Ilies, L.; Verma, P.; Ichikawa, S.; Nakamura, E. *Chem. Lett.* **2014**, 43, 726.
- (10) (a) Braun, T.; Perutz, R. N.; Sladek, M. I. *Chem. Commun.* **2001**, 2254. (b) Braun, T.; Izundu, J.; Steffen, A.; Neumann, B.; Stammler, H.-G. *Dalton Trans.* **2006**, 5118.
- (11) Jin, G.; Zhang, X.; Cao, S. *Org. Lett.* **2013**, 15, 3114.
- (12) Kiso, Y.; Tamao, K.; Kumada, M. *J. Organomet. Chem.* **1973**, 50, C12–C14.
- (13) Tamao, K.; Sumitani, K.; Kiso, Y.; Zembayashi, M.; Fujioka, A.; Kodama, S. I.; Nakajima, I.; Minato, A.; Kumada, M. *Bull. Chem. Soc. Jpn.* **1976**, 49, 1958.
- (14) (a) Guo, H.; Kong, F.; Kanno, K.-I.; He, J.; Nakajima, K.; Takahashi, T. *Organometallics* **2006**, 25, 2045. (b) Yang, X.; Sun, H.; Zhang, S.; Li, X. *J. Organomet. Chem.* **2013**, 723, 36. (c) Sun, A. D.; Leung, K.; Restivo, A. D.; LaBerge, N. A.; Takasaki, H.; Love, J. A. *Chem. - Eur. J.* **2014**, 20, 3162. (d) Xiao, S.-H.; Xiong, Y.; Zhang, X.-X.; Cao, S. *Tetrahedron* **2014**, 70, 4405. (e) Yu, D.; Wang, C.-S.; Yao, C.; Shen, Q.; Lu, L. *Org. Lett.* **2014**, 16, 5544. For the coupling of aryl fluorides with diarylmethanes, see: (f) Li, J.; Wu, C.; Zhou, B.; Walsh, P. J. *J. Org. Chem.* **2018**, 83, 2993. For a multicomponent reaction leading to Csp²–Csp³ bond formation, see: (g) Iwasaki, T.; Fukuoka, A.; Min, X.; Yokoyama, W.; Kuniyasu, H.; Kambe, N. *Org. Lett.* **2016**, 18, 4868. During the preparation of this manuscript, a Ni-catalyzed protocol for the coupling of aryl fluorides with secondary Grignard reagents appeared: (h) O'Neill, M. J.; Riesebeck, T.; Cornella, J. *Angew. Chem., Int. Ed.* **2018**, 57, 9103–9107.
- (15) For the use of LiCH₂SiMe₃, in coupling reactions, see: (a) Wang, Z.-L.; Zhao, L.; Wang, M.-X. *Chem. Commun.* **2012**, 48, 9418. (b) Giannerini, M.; Fañanás-Mastral, M.; Feringa, B. L. *Nat. Chem.* **2013**, 5, 667. (c) Leiendecker, M.; Hsiao, C.-C.; Guo, L.; Alandini, N.; Rueping, M. *Angew. Chem., Int. Ed.* **2014**, 53, 12912. (d) Heijnen, D.; Hornillos, V.; Corbet, B. P.; Giannerini, M.; Feringa, B. L. *Org. Lett.* **2015**, 17, 2262. (e) Jia, Z.; Liu, Q.; Peng, X.-S.; Wong, H. N. C. *Nat. Commun.* **2016**, 7, 10614.
- (16) Muto, K.; Yamaguchi, J.; Lei, A.; Itami, K. *J. Am. Chem. Soc.* **2013**, 135, 16384.